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Mechanical Engineering & Computer Science — Engineering Automation & Software
Development

Professional Portfolio — Selected Engineering Projects

3D AHU App

Interactive 3D reconstruction and parametric visualization system for large HVAC units

Problem

Large Air Handling Units (AHUs) are typically documented through static drawings, photographs, and fragmented CAD files. This makes spatial reasoning, configuration comparison, and communication between engineering teams and clients inefficient and error-prone.

Solution

Designed and developed a web-based interactive application that reconstructs full-scale AHUs using photogrammetry and overlays parametric logic for inspection, exploration, and comparison. The system enables explodable views, sectional inspection, and side-by-side condition comparisons directly in the browser.

My Role

- Led end-to-end system design and development.
- Designed the photogrammetry pipeline and data preprocessing workflow.
- Implemented Three.js scene architecture and interaction logic.
- Built parametric abstraction on top of scanned geometry.

Technical Depth

- Photogrammetry processing, mesh cleanup, and spatial alignment.
- Hybrid modeling approach combining scan fidelity with parametric logic.
- Three.js scene management, interaction handling, and performance optimization.
- Implementation of exploded views, component isolation, and comparison workflows.

Impact

- Enabled engineers and clients to interactively explore complex AHU systems.
- Improved design clarity and reduced communication friction during reviews.
- Established a scalable foundation for digital-twin-style workflows.

QuickScope

Full-stack estimation and automation platform for multi-department engineering workflows

Problem

Project estimation workflows were slow, manual, and fragmented across multiple tools. Data inconsistencies and repetitive manual steps increased cycle time and reduced reliability.

Solution

Built QuickScope, a full-stack automation platform that integrates engineering, sales, and reporting workflows into a single system. The platform automates metadata handling, reporting, and estimation logic while maintaining compatibility with existing enterprise tools.

My Role

- Designed system architecture and automation pipelines.
- Implemented integrations across C#, VBA, Salesforce, Excel, and Google Maps API.
- Developed UI shortcuts and workflow optimizations for speed and usability.

Technical Depth

- Cross-platform data pipelines and synchronization.
- API integrations and automated metadata propagation.
- VBA and C# automation logic for reporting and estimation.
- UI/UX decisions focused on minimizing user friction.

Impact

- Reduced estimation cycle time by approximately 45%.
- Streamlined reporting across departments.
- Improved usability and adoption of estimation tools.

Autonomous Sorting Machine

Automated bearing sorting and packaging system combining mechanical design, vision, and ML

Problem

Manual sorting and packaging of small mechanical components was inefficient and prone to classification errors.

Solution

Designed and built an autonomous sorting system integrating mechanical hardware, computer vision, and machine learning. The system identifies components using a vision model and routes them through a mechanically actuated sorting and packaging process.

My Role

- Designed mechanical subsystems and physical layout.
- Integrated Raspberry Pi-based vision system.
- Coordinated ML classification output with actuation and control logic.

Technical Depth

- Gravity-fed hopper design with custom sieves and 3D-printed components.
- Vision pipeline using Raspberry Pi and Teachable Machine.
- Timing and coordination between sensing, classification, and mechanical actuation.

Outcome

- Delivered a functional prototype demonstrating end-to-end automation.
- Validated integration of mechanical design, ML inference, and physical control.

Digitally Manufactured Reactor

Fully CAD-driven design and fabrication of a multi-process mechanical assembly

Scope

Designed and fabricated a digitally manufactured reactor using a fully CAD-driven workflow and multiple manufacturing processes.

System Overview

The assembly consists of:

- A 3D-printed reactor body.
- A laser-cut acrylic enclosure.
- Machined components including an aluminum plate and ceramic tube.

Engineering Focus

- Design-for-manufacturing and design-for-assembly decisions.
- Translation of CAD models into fabrication-ready files.

Outcome

- Delivered a physical, functional assembly.
- Demonstrated integration of CAD modeling, additive manufacturing, laser cutting, and purchased components into a cohesive system.

Audio Feature Analysis & Text-to-Speech

Exploration of audio feature similarity and ML-based speech synthesis

Overview

Developed an experimental tool focused on audio feature extraction, comparison, and text-to-speech synthesis.

Technical Stack

- Librosa for audio feature extraction and similarity analysis.
- YourTTS for text-to-speech generation.
- Streamlit for a lightweight interactive front-end.

What This Demonstrates

- Audio signal processing and feature engineering.
- Integration of ML models into user-facing applications.
- Rapid prototyping of ML-driven tools.

Additional Projects

Solar Car Linkage System

- Designed a deployment linkage for a solar array (0° driving $\rightarrow 90^\circ$ charging).
- Developed MATLAB scripts to verify Grashof conditions and degrees of freedom.
- Modeled, scaled, and validated the mechanism in CAD.
- Built and tested a physical prototype.

EcoTrail AI

- Natural language trip parsing.
- Live Google Maps routing and interactive path visualization.
- CO₂ emission modeling using EPA datasets.
- Behavior analytics and personalized feedback.

Data Analysis & Engineering Economy

- Monte Carlo simulation (10,000 ownership paths).
- CVaR, stochastic dominance, and Bayesian updating.
- VBA-driven Excel models for MIRR, IRR, depreciation, and cash-flow analysis.

This portfolio highlights selected projects demonstrating systems thinking, technical depth, and engineering judgment across mechanical, software, and data-driven domains.